OENG1209: Engineering Science

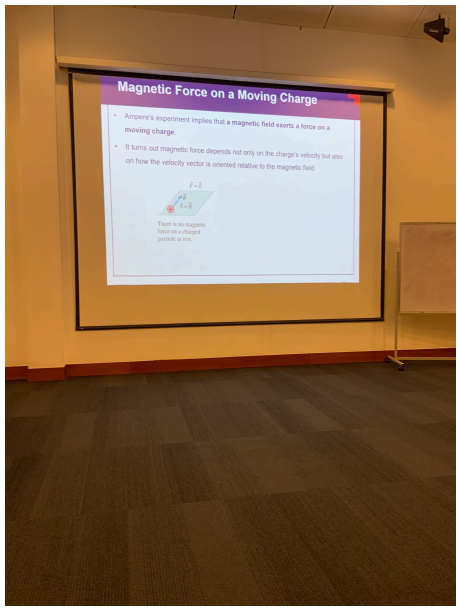
Week 10:

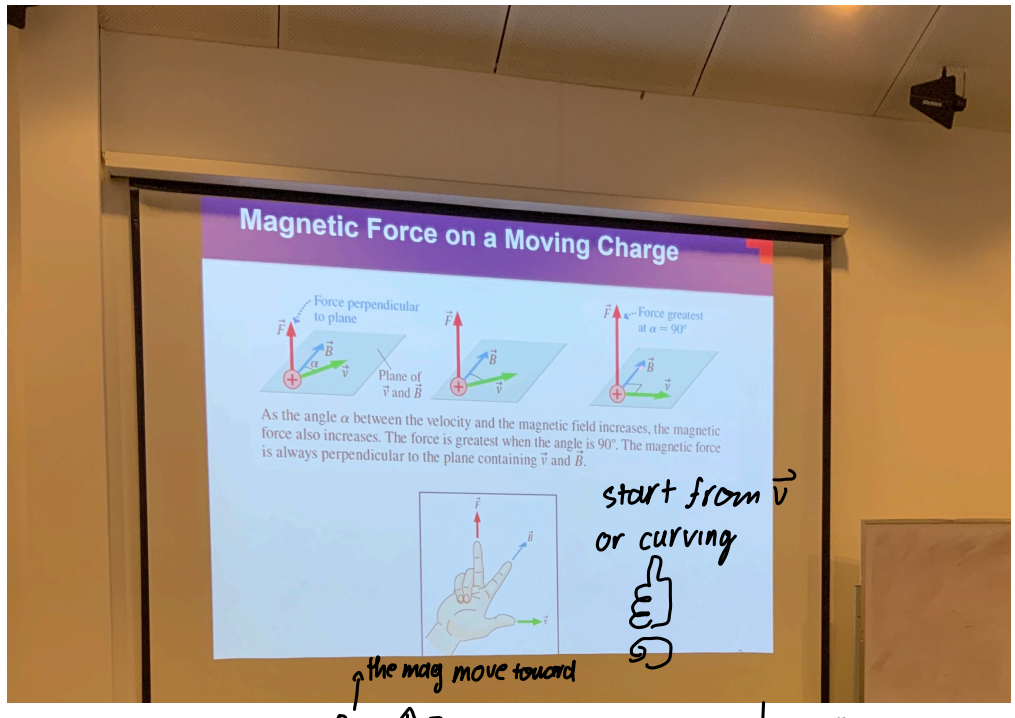
Electromagnetic Induction

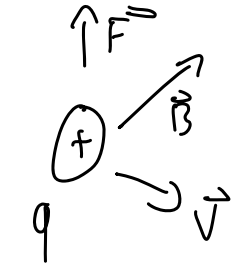
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Today's Topics

- Solenoid
- Electromagnetic Induction
- Magnetic Flux
- Faraday's Law and Lenz's Law



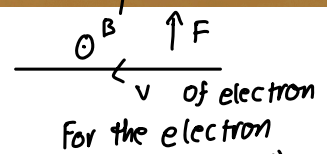




$\vec{F} = q\vec{v} \times \vec{B}$

start from $\vec{v}$ or curving

the mag move toward

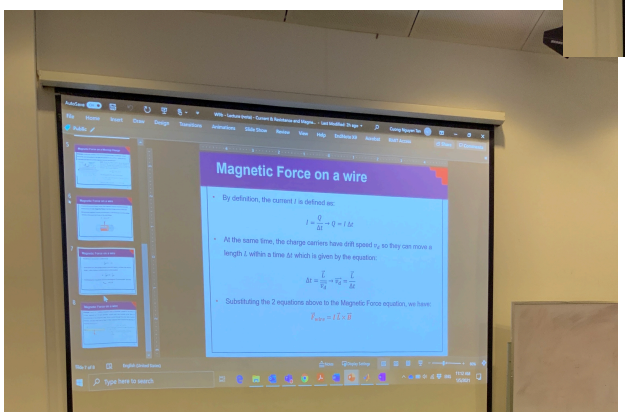


For the electron

$\vec{F} = q\vec{v} \times \vec{B}$

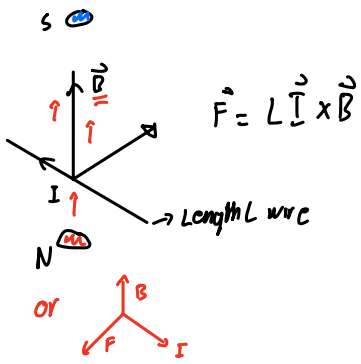


$\vec{F} = I\vec{L} \times \vec{B}$



$F_{wire} = I\vec{L} \times \vec{B}$

$= L\vec{I} \times \vec{B}$



$$\vec{F} = L\vec{I} \times \vec{B}$$

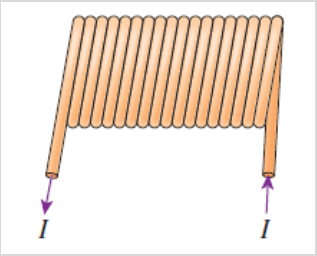
Solenoid

An electromagnet made from a coil of wire

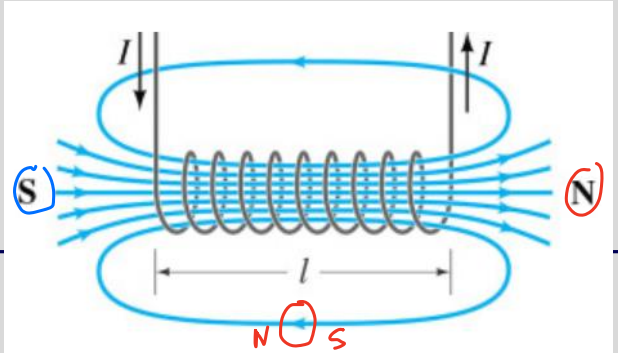
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Solenoid

- The **N-turn coil** with a current running through is a special device that is quite popular in engineering design. It is called **a solenoid**.



- **A solenoid** will generate a uniform magnetic field when a current is ran through it. Thus, a solenoid is often called **an electromagnet**.





Solenoid

- **The strength of the uniform magnetic field** inside a solenoid, generated by the **current** I is (proof available upon request):

$$B_{solenoid} = \frac{\mu_0 NI}{L}$$

$\mu_0 =$ the permeability constant of vacuum.

$N =$ the number of turns on the coil.

$L =$ the length of the coil.

$I =$ the current running through the coil.

Electromagnetic Induction

Generating current from a magnetic field

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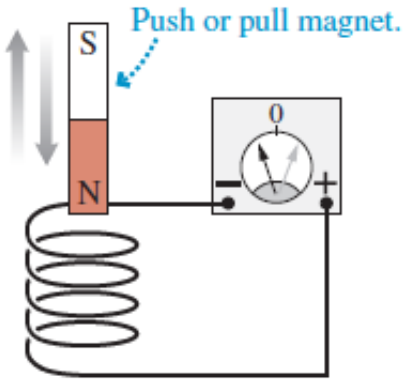
Electromagnetic Induction

Discovery

In 1831, Faraday found that there is **a current** in a coil of wire if and only if the **magnetic field passing through the coil is changing** → New way to generate a current → **induced current**

strength of current depend on N and velocity

He pushed a bar magnet into a coil of wire. This action caused a momentary deflection of the current-meter needle, although *holding* the magnet inside the coil had no effect. A quick withdrawal of the magnet deflected the needle in the other direction.

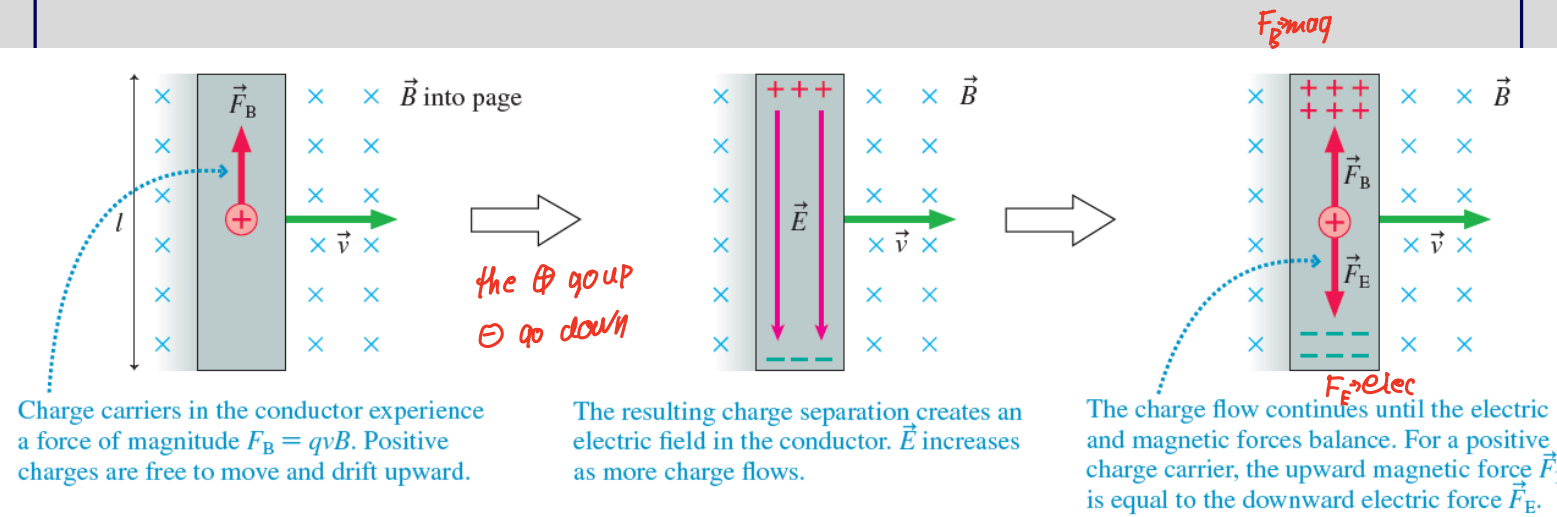


Pushing the magnet into the coil or pulling it out creates a momentary current.

Electromagnetic Induction

Motional emf

- What happens if the magnetic is fixed and the current moves?
- Consider a conductor of **length** l that moves with **velocity** $\vec{v}$



Charge carriers in the conductor experience a force of magnitude $F_B = qvB$. Positive charges are free to move and drift upward.

The resulting charge separation creates an electric field in the conductor. $\vec{E}$ increases as more charge flows.

The charge flow continues until the electric and magnetic forces balance. For a positive charge carrier, the upward magnetic force $\vec{F}_B$ is equal to the downward electric force $\vec{F}_E$.

Electromagnetic Induction

Motional emf

- Consider a **conductor of length l** that moves with **velocity $\vec{v}$,** hence they each experience a magnetic force:

$$|\vec{F}_B| = q\ |\vec{v}||\vec{B}|$$

- The force causes the charge carriers to move, separating the positive and negative charges. They then create an electric field inside the conductor with force:

$$F_E = qE$$

- F_E will continue to grow until the system is in equilibrium which is:

$$E = vB$$

Electromagnetic Inductio
Motional emf

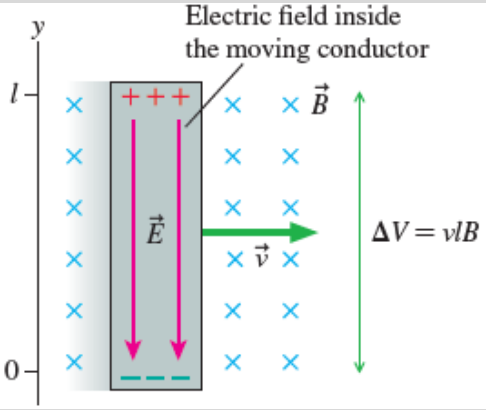
- **The electric field** will create an **electric potential difference** between the two ends of the moving conductor which is:

$$\Delta V = -\int_0^L E_y dy = -\int_0^L (-vB) dy = vLB$$

- This means the **motion of the wire** through a magnetic field induces a **potential difference** vLB between the ends of the conductor.
- This potential difference is called **the motional electromotive force (emf)** $\mathcal{E}$

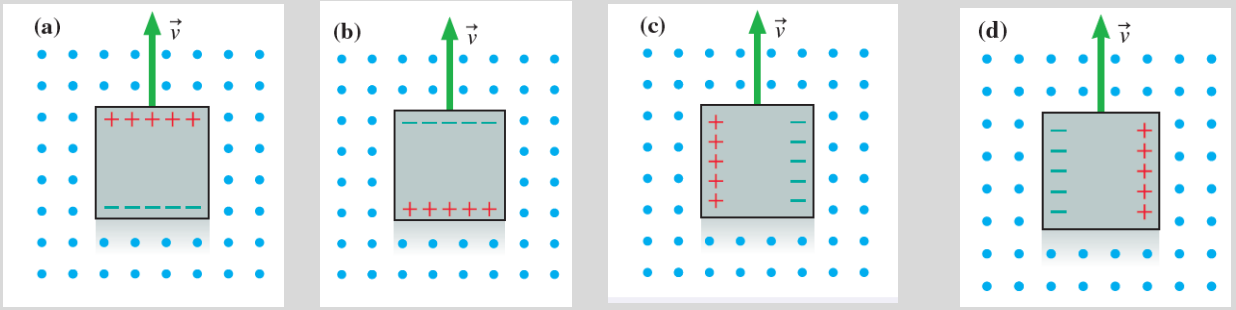
$$\mathcal{E} = vLB$$

the diff in potential



Electromagnetic Induction

Stop to think: A square conductor moves through a uniform magnetic field. Which of the figures shows the correct charge distribution on the conductor?



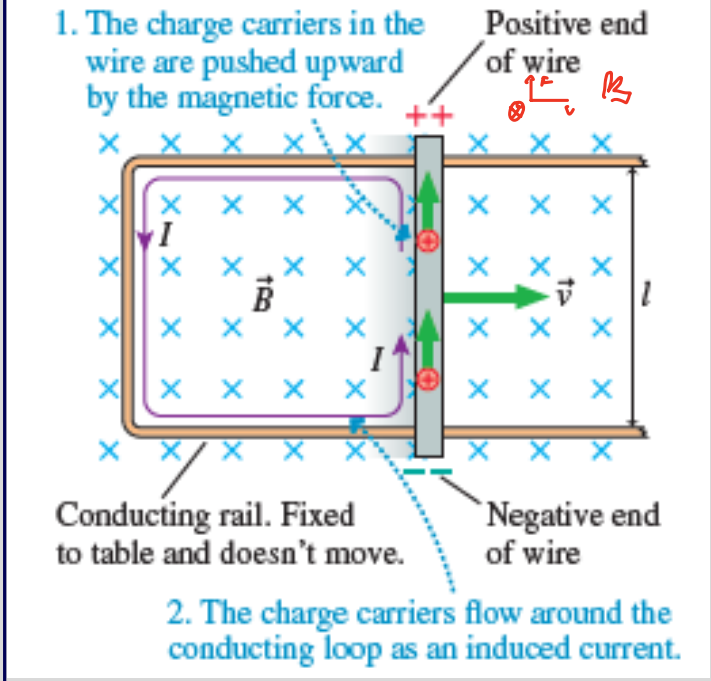
Electromagnetic Induction - Motional emf

- We include the inductor in a circuit, the rail cannot move.

→ **an induced current** (**counterclockwise**).

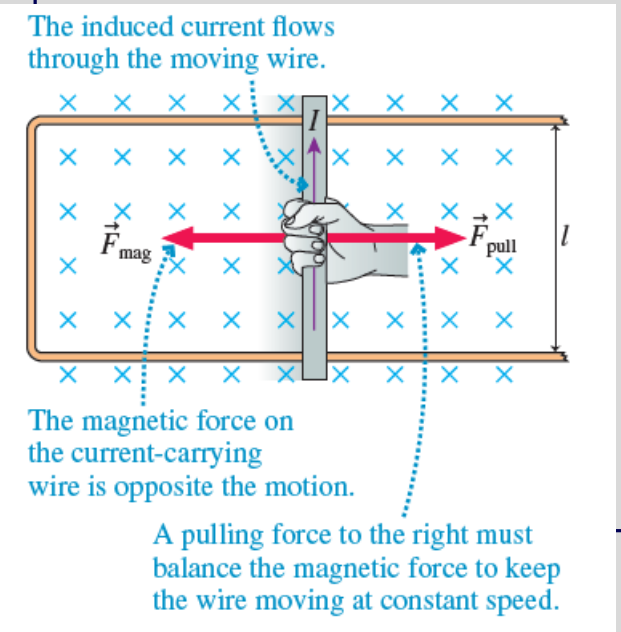
- If the total resistance of the circuit is R, the induced current is given by Ohm's law as:

$$I = \frac{\mathcal{E}}{R} = \frac{vLB}{R}$$



Electromagnetic Induction
Motional emf

- **Question**: If we bring this circuit to outer space and give the wire a push, will it keep going forever?
- According to the equation $\vec{F}_{wire} = I\,\vec{L} \times \vec{B}$, there will be **a magnetic force** pushing the wire to the left as it moves to the right to generate a current.



Hence, if we want to keep the wire moving, we need to apply a Force:

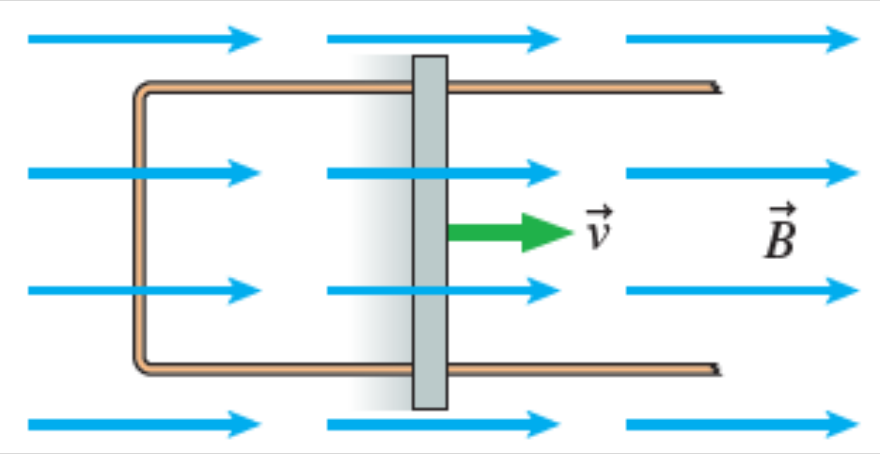
$$F_{pull} = F_{mag} = I\,L\,B = \frac{(vLB)}{R}LB = \frac{vL^2B^2}{R}$$

Recall that **Power** is the rate at which work is done on the wire such that:

$$P_{input} = F_{pull}v = \frac{v^2L^2B^2}{R}$$

Electromagnetic Induction

Stop to think: Is there an induced current in the circuit?



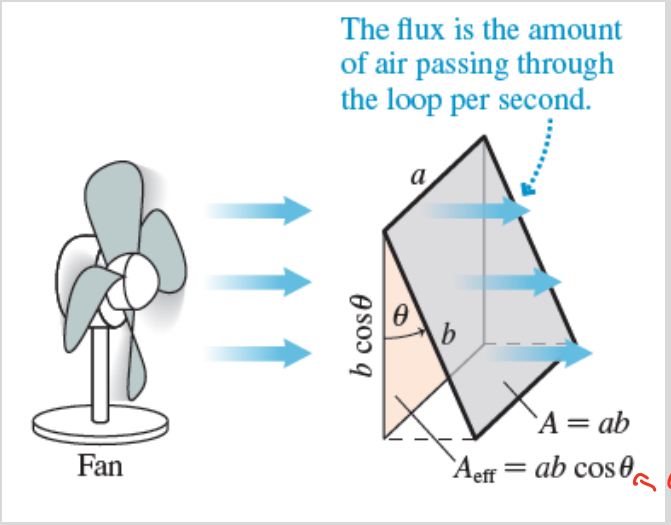
Magnetic Flux

Generating current from a magnetic field

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Magnetic Flux

- The flux of the Magnetic field which → how much of the magnetic field flows through our loop
- Imagine holding a **rectangular loop** in front of a fan. The **amount of air** flowing through the loop is called **the flux**, and it is dependent on the angle of the loop.



The flow is maximum when $\theta = 0$ and is zero when $\theta = 90^o$.

Hence, we define the effective area of the loop as:

$$A_{eff} = ab\cos(\theta) = A\cos(\theta)$$

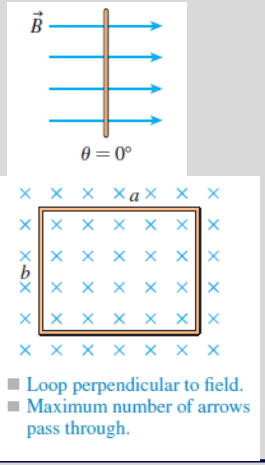
0 → 100% eff

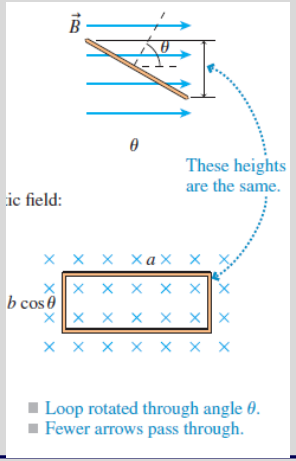
Magnetic Flux

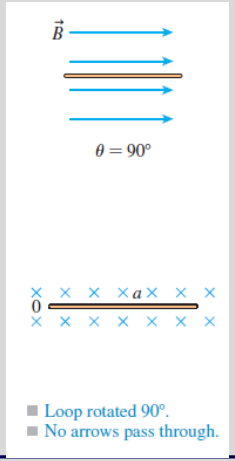
- We can apply this idea to a **magnetic field** passing through a loop and replace the air with the amount of arrows representing the magnetic field. The **number of arrow** passing through the loop will depend on **2 factors**:
  - **The density of arrow**, which is proportional to $|\vec{B}|$.
  - **The effective area** $A_{eff} = A\cos\theta$ of the loop.

↑ area → ↑ flux
θ ↑ → ↓ flux
θ = 0 => max flux

$$\Phi = \int \vec{B} \cdot d\vec{A}$$







Magnetic Flux

- Thus, we can define **the magnetic flux Φ_m** as:

$$\Phi_m = A_{eff}|\vec{B}| = A|\vec{B}|cos\theta$$

- The unit of the Magnetic Flux is **Weber**:

$$1\ weber = 1\ Wb = 1\ Tm^2$$

- Recall the definition of **a Dot Product** between 2 vectors $\vec{a}$ and $\vec{b}$:

$$\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|\cos(\theta)$$

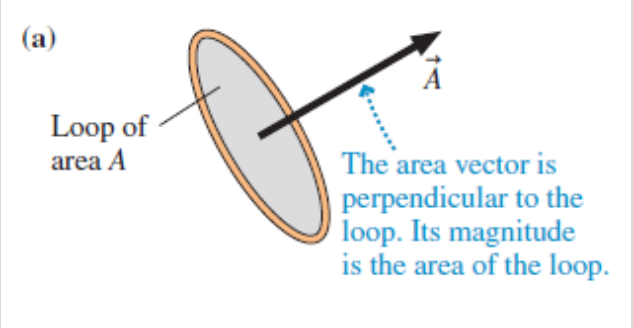
- Hence, we can write the definition of the magnetic flux as a dot product:

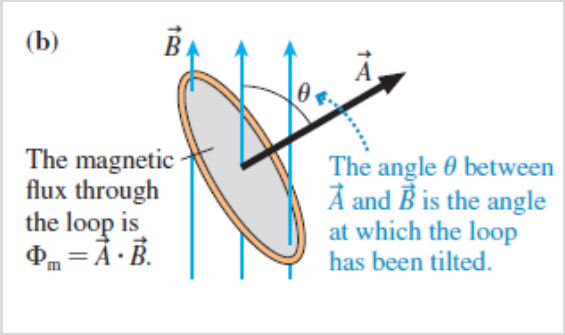
$$\Phi_m = \vec{A} \cdot \vec{B} = |\vec{A}||\vec{B}|cos(\theta)$$

What is vector $\vec{A}$?

Magnetic Flux

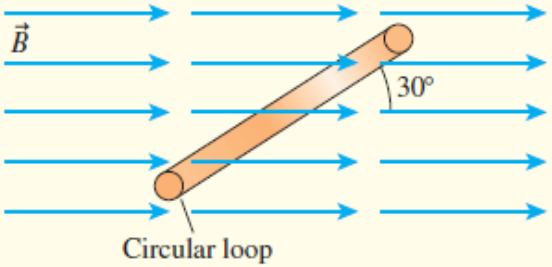
- The **vector** $\vec{A}$ in the previous equation is called **the area vector**.
- The **area vector** $\vec{A}$ is defined to be the vector perpendicular to the loop, with **magnitude equal to the area** A of the loop.
  - The unit of vector $\vec{A}$ is m^2.





Magnetic Flux

Example: The figure below shows an edge view of a 10 cm diameter circular loop in a uniform 0.050 T magnetic field. What is the magnetic flux through the loop?



SOLVE Angle θ is the angle between the loop's area vector $\vec{A}$, which is perpendicular to the plane of the loop, and the magnetic field $\vec{B}$. In this case, $\theta = 60°$, not the 30° angle shown in the figure. Vector $\vec{A}$ has magnitude $A = \pi r^2 = 7.85 \times 10^{-3}\ \text{m}^2$. Thus the magnetic flux is

$$\Phi_m = \vec{A} \cdot \vec{B} = AB\cos\theta = 2.0 \times 10^{-4}\ \text{Wb}$$

Lenz's Law&
Faraday's Law

Quantifying the emf

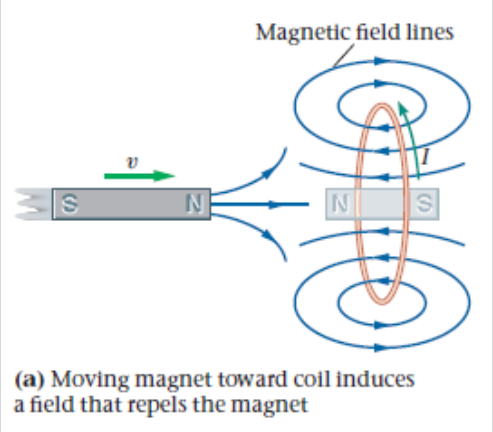
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Lenz's Law

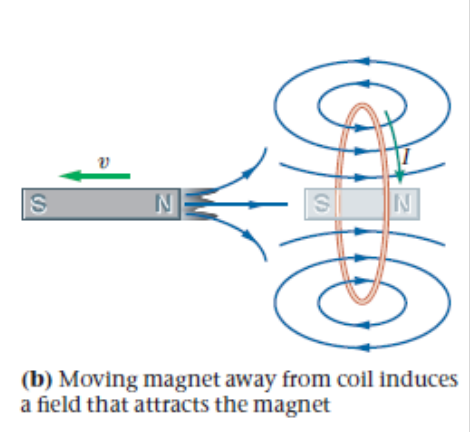
- **Lenz's law → determine the dirciton of the induced current**
- When there is an induced current in a closed, conducting loop, **the direction of the induced current** is such that **the induced magnetic field** opposes the change in the flux.
- In other words, an induced current always flows in a direction that opposes the change that cause it.

Lenz's Law

- The "**change**" can happen in three ways:
  - **The Magnetic field** through the loop changes (increases or decreases).
  - **The loop changes** in area or angle.
  - **The loop moves** into or out of a magnetic field.



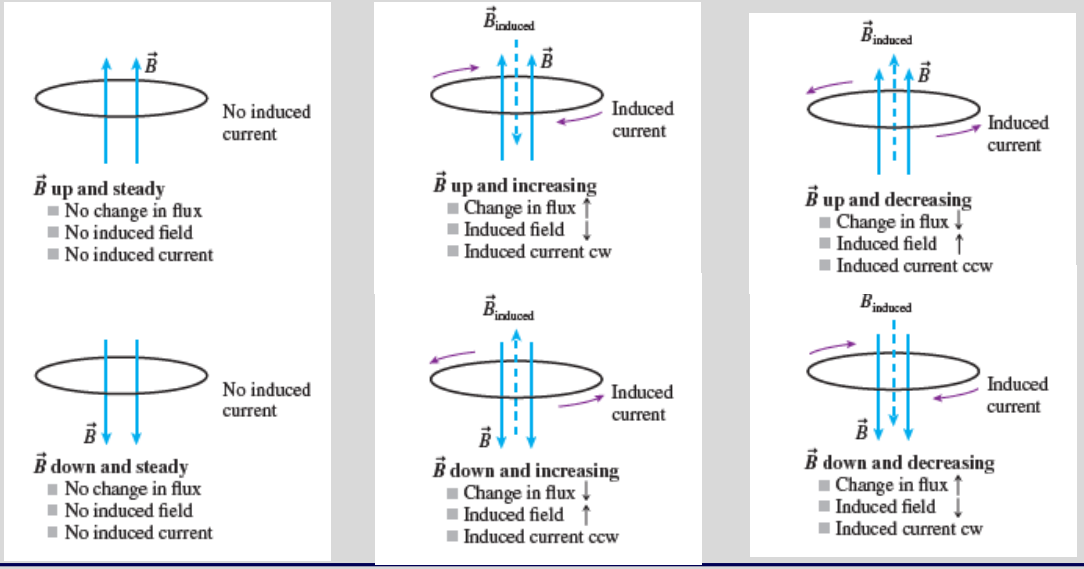
(a) Moving magnet toward coil induces a field that repels the magnet



(b) Moving magnet away from coil induces a field that attracts the magnet

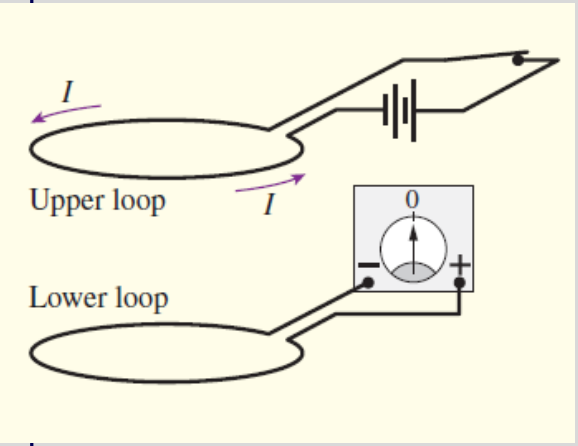
Lenz's Law

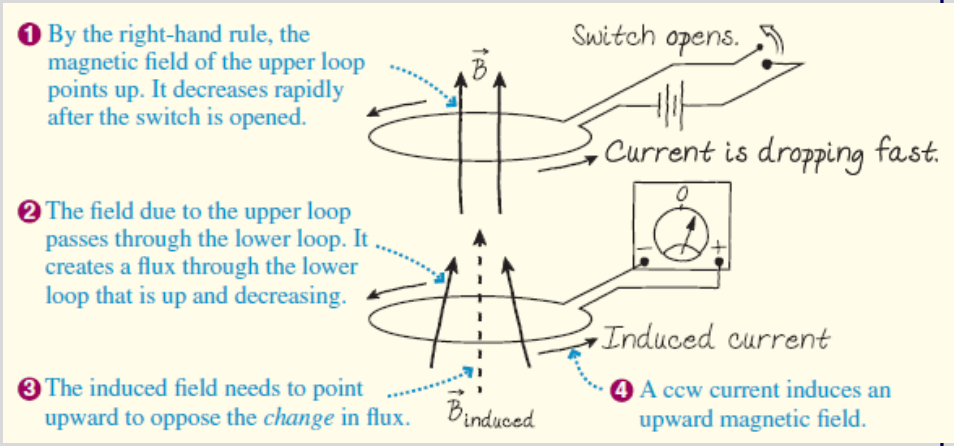
- There are **6 basic situations** for the application of **Lenz's Law**. The magnetic field can point either up or down through the loop.
- For each case, **the flux** can either increase, hold steady, or decrease in strength.



Lenz's Law

Example: The figure below shows two loops, one above the other. The upper loop has a battery and a switch that has been closed for a long time how does the lower loop respond when the switch is opened in the upper loop?





Faraday's Law

- This emf associated with a **changing magnetic flux** is called **an induced emf** $\mathcal{E}$. Then, if there is a complete circuit having **resistance** R, then the induced current is:

$$I_{induced} = \frac{\mathcal{E}}{R} \; (direction\ by\ Lenz's\ law)$$

- **Faraday's Law** states that an **emf** $\mathcal{E}$ is induced around a closed loop if **the magnetic flux** through the loop changes. The magnitude of the emf is given by:

↑change in mag flux

$$\mathcal{E} = \left| \frac{d\Phi_m}{dt} \right|$$

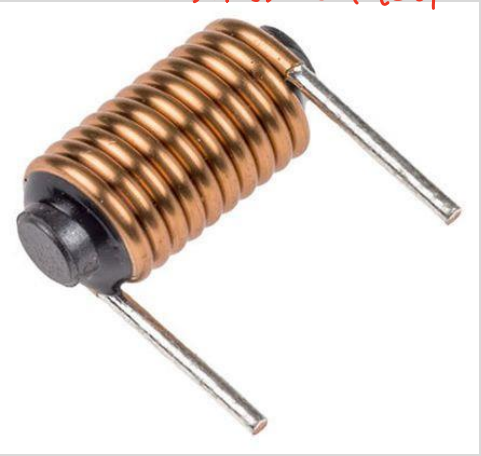
- In other words, the induced emf is the rate of change of the magnetic flux through the loop.

Faraday's Law

- An N-turn coil of wire in a changing magnetic field acts like N batteries in series. Thus the **induced emf** of the entire coil is:

$$\mathcal{E} = N \left| \frac{d\Phi_{per\ coil}}{dt} \right|$$

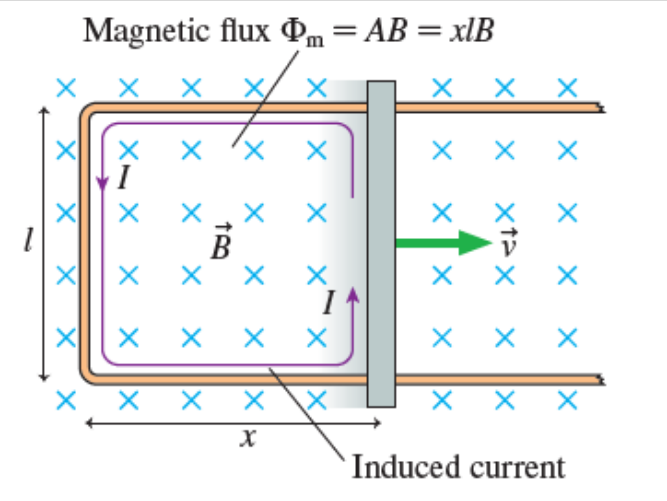
N: number of coil loop



Faraday's Law

Re-exame **the generator circuit** seen previously and apply **Faraday's Law.**

Goal: compute the induced current in terms of v, L, B and R



- Notice as the right wire is moved to the right, the magnetic flux through the loop is also increasing ($\vec{A}$ is increasing).
- The magnetic flux is given by:

$$\Phi_m = |\vec{A}||\vec{B}|cos90^o = AB = xLB$$

- According to Faraday's law, the induce emf is:

$$\mathcal{E} = \left|\frac{d\Phi_m}{dt}\right| = \frac{dx}{dt}LB = vLB$$

- Thus the induce current is:

$$I = \frac{\mathcal{E}}{R} = \frac{vLB}{R}$$

Faraday's Law

Example: A conducting loop is halfway into a magnetic field. Suppose the magnetic field begins to increase rapidly in strength. What happens to the loop?

a) The loop is pushed upward.

b) The loop is pushed downward.

c) The loop is pulled to the left.

d) The loop is pushed to the right

e) The loop does not move.

